Held-out mental health versus CNIO

Same VQ-VAE a0 and the same 6-topic LDA. Recorded days only. Mental health is held out of training.

Mental health

1,358 patients
370,841 days
emotion_inter or emotion_intra

CNIO oncology

175 patients
51,972 days
DB1 PMP

Where the two distances are measured

Raw day → Encoder continuous z → Nearest of 256 codes → Decoder reconstruction

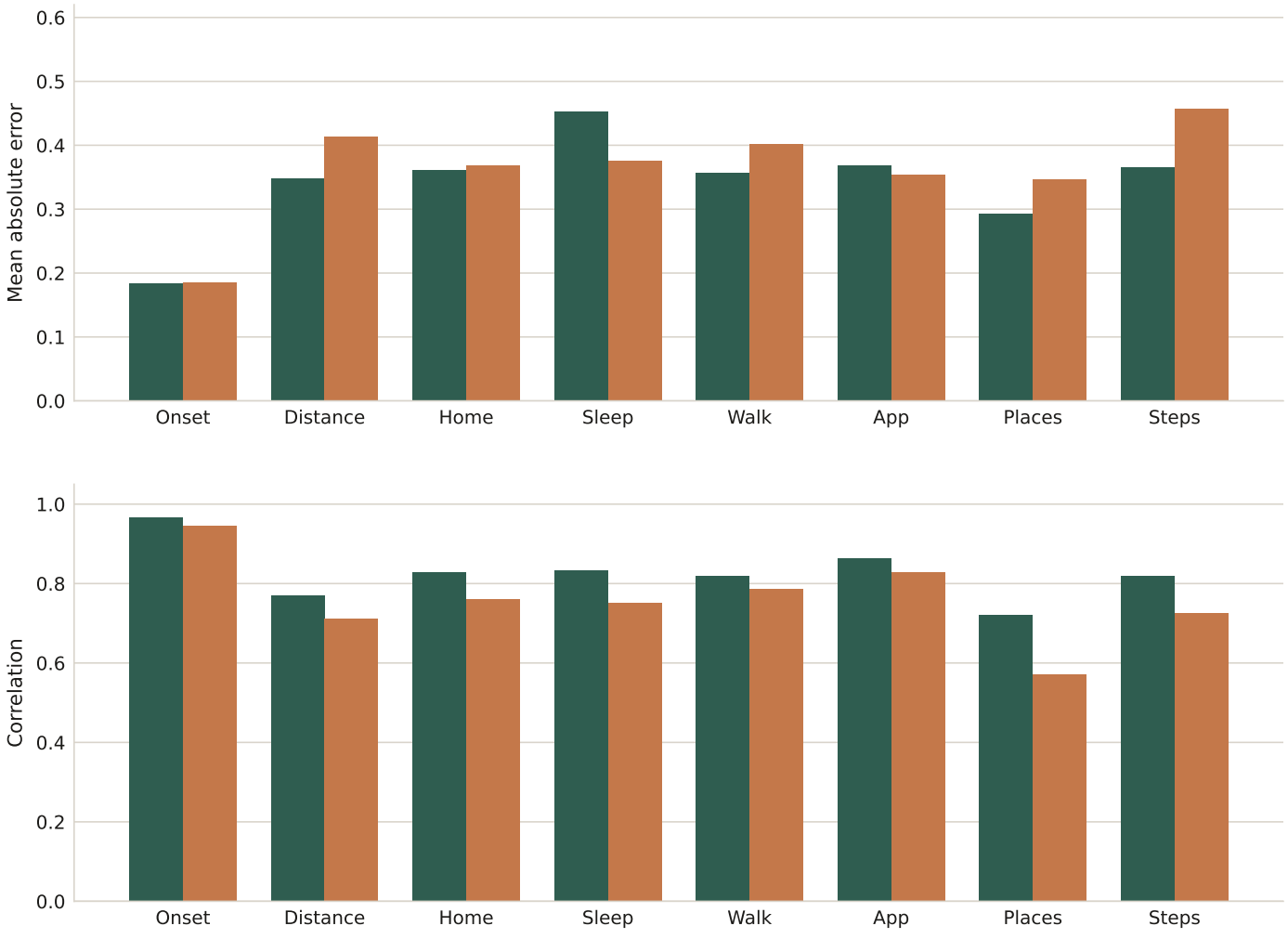
latent distance to the code

error against the raw day

Reconstruction, scaled training space

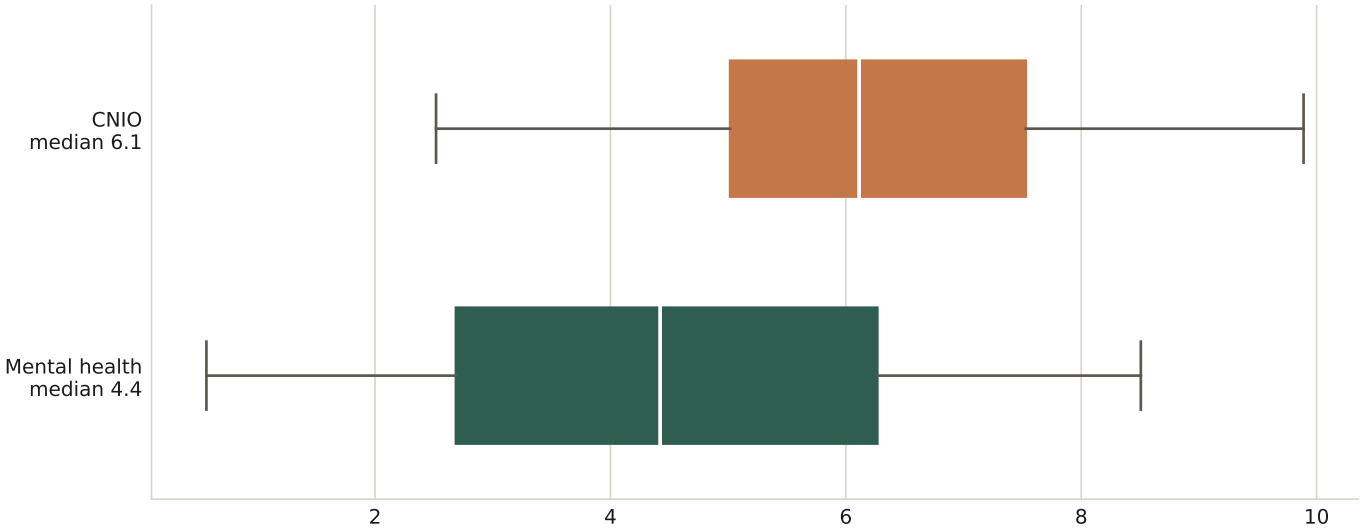
Mental health CNIO

Pooled correlation 0.83 in mental health and 0.77 in CNIO. Weekend excluded.



One code, with a margin, and a residual

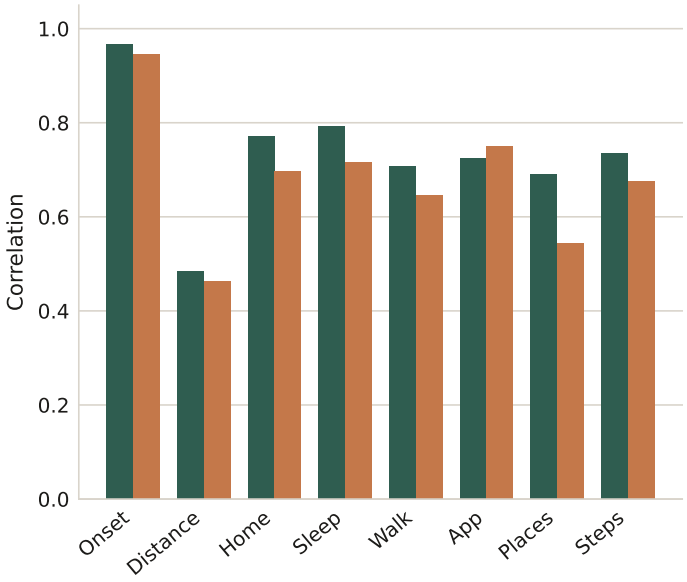
Distance is Euclidean, from the continuous embedding to the nearest code.
The residual is the decoded day versus the raw day, back in original units.

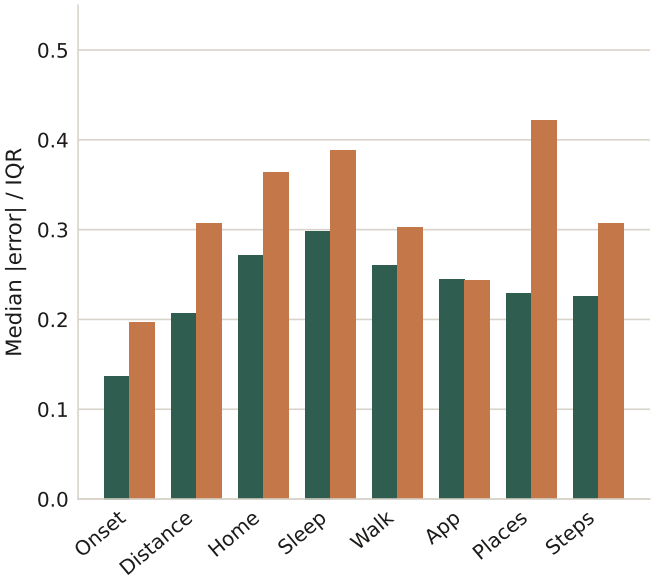


Whiskers 5th–95th. Day-level median distance 4.0 and 6.2. Margin over the second code 0.82 and 0.98.

Original units

Left: correlation with the raw day. Right: median error as a fraction of the IQR.



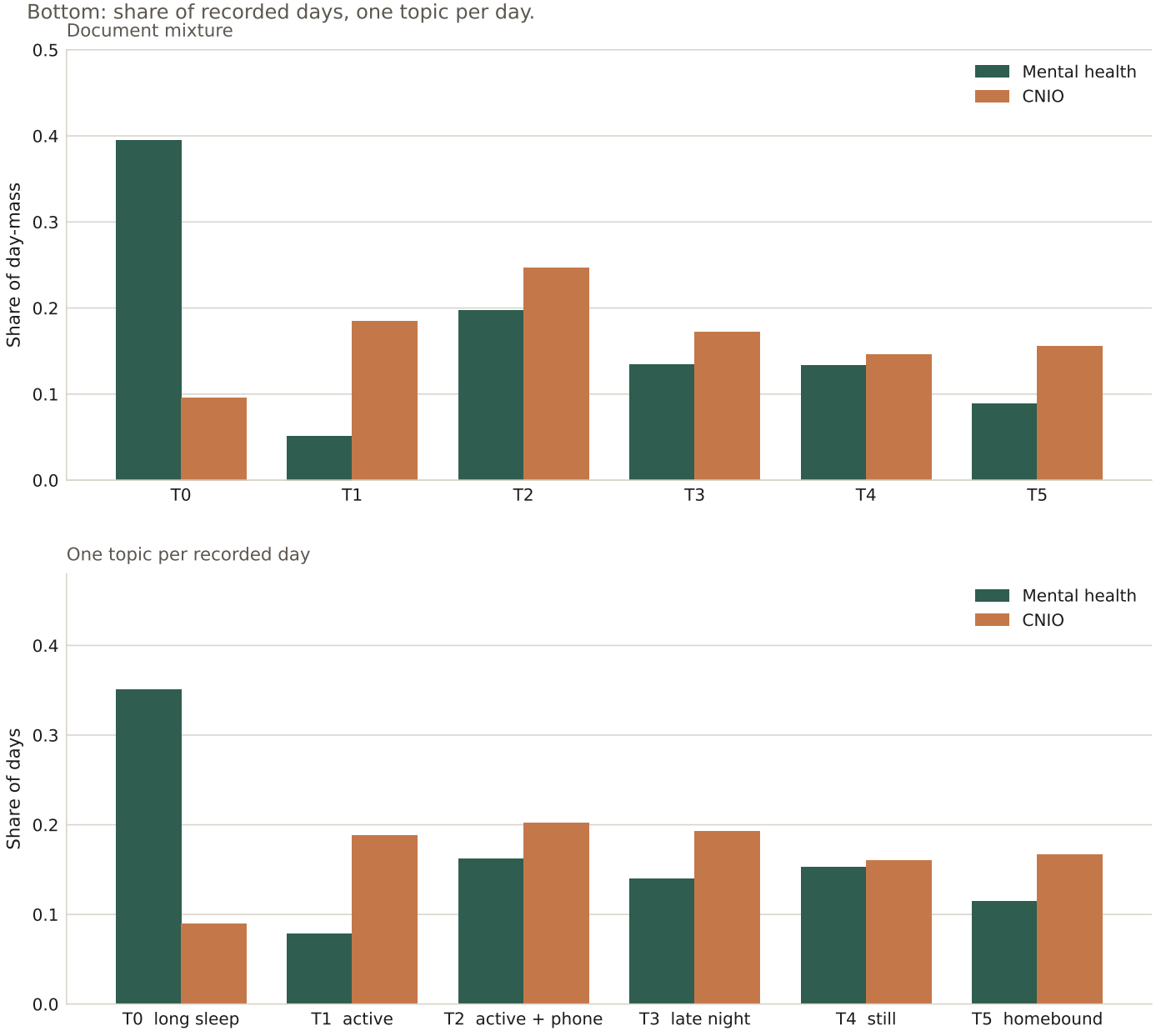


Median absolute error, original units

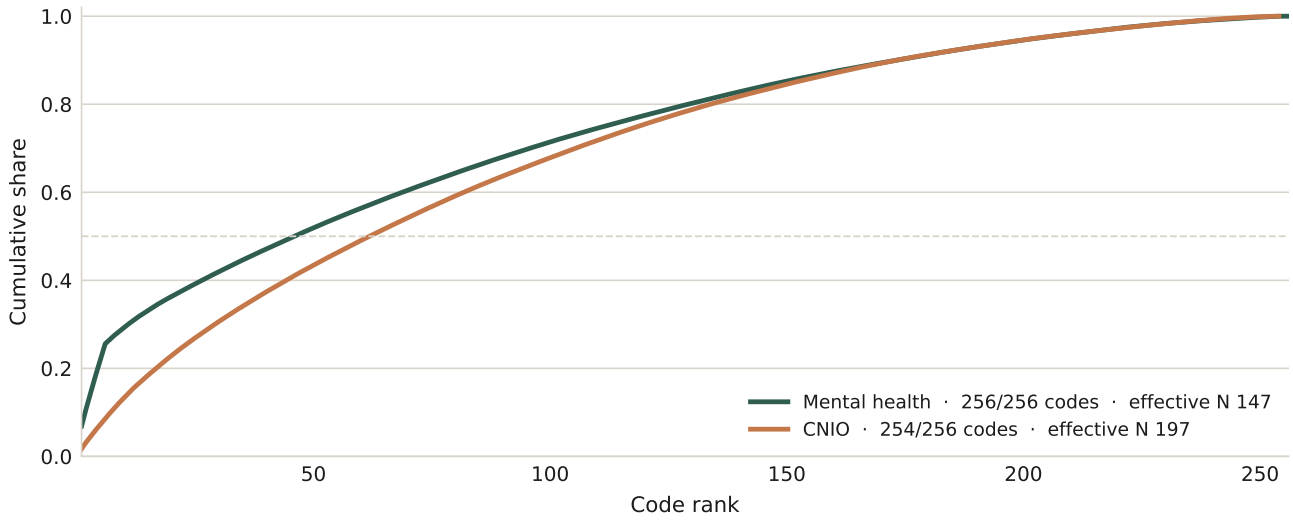
Feature	Sleep onset	Sleep length	Steps	Walking	App	Home	Distance	Places
Mental	24 min	1.1 h	1,377	15 min	54 min	2.7 h	4.5 km	0.7
CNIO	24 min	0.9 h	2,399	18 min	39 min	3.1 h	4.5 km	0.8

The mixtures differ. The codebook does not collapse.

Top: day-weighted topic mixture of each patient (JS 0.076).
Bottom: share of recorded days, one topic per day.

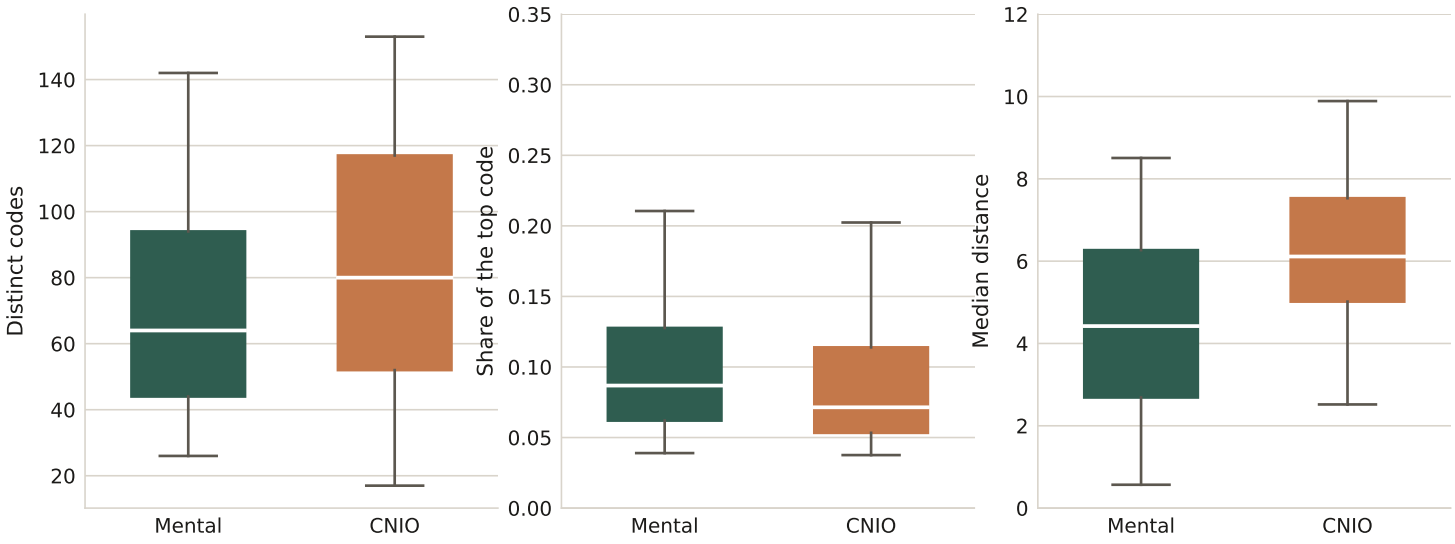


Cumulative share of days, codes ranked from most to least used



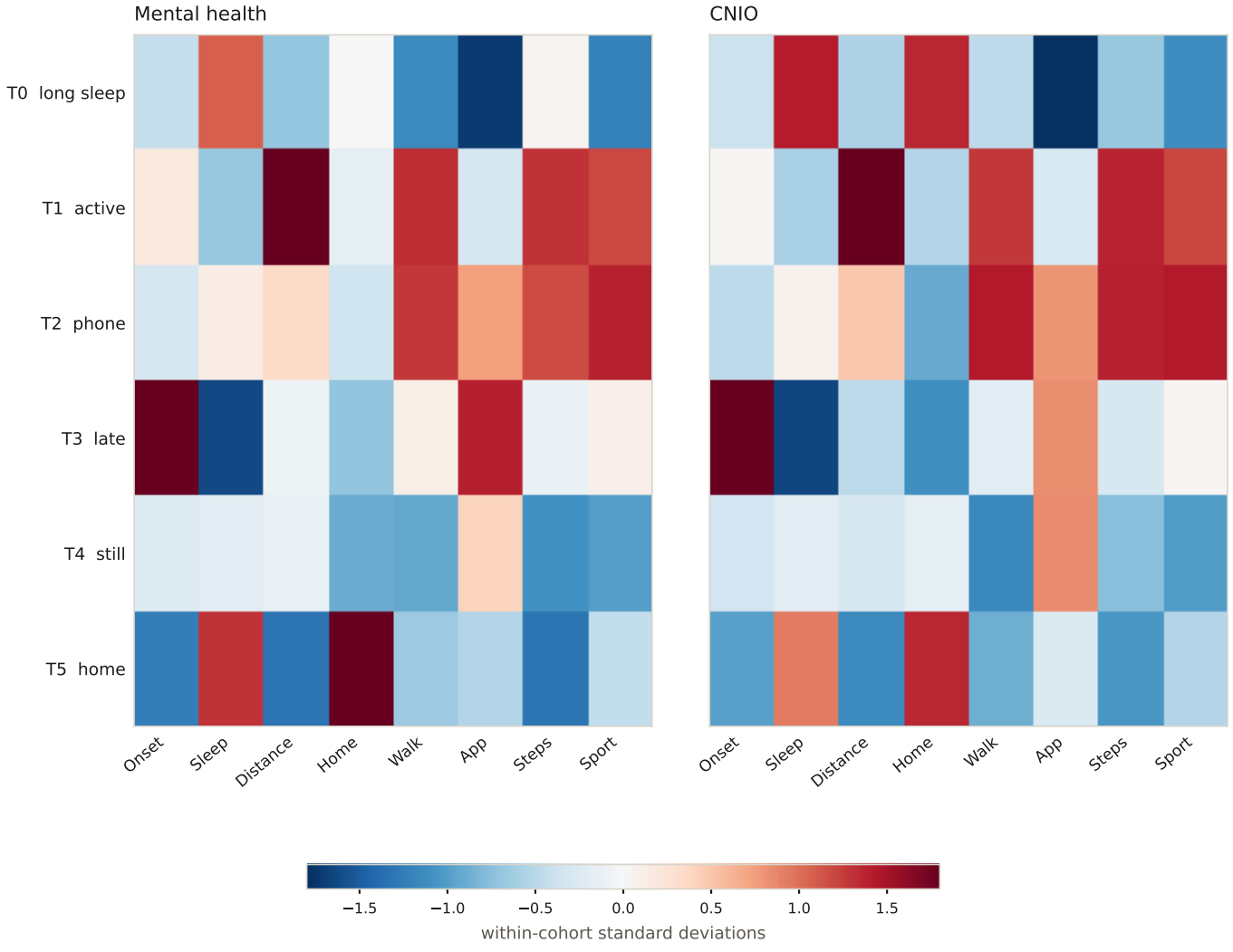
Dozens of codes, none of them the whole patient

Median patient: 64 codes and 9% on the top code in mental health; 80 codes and 7% in CNIO.
Dominant topic mass 0.59 and 0.58. Whiskers are the 5th and 95th percentiles.



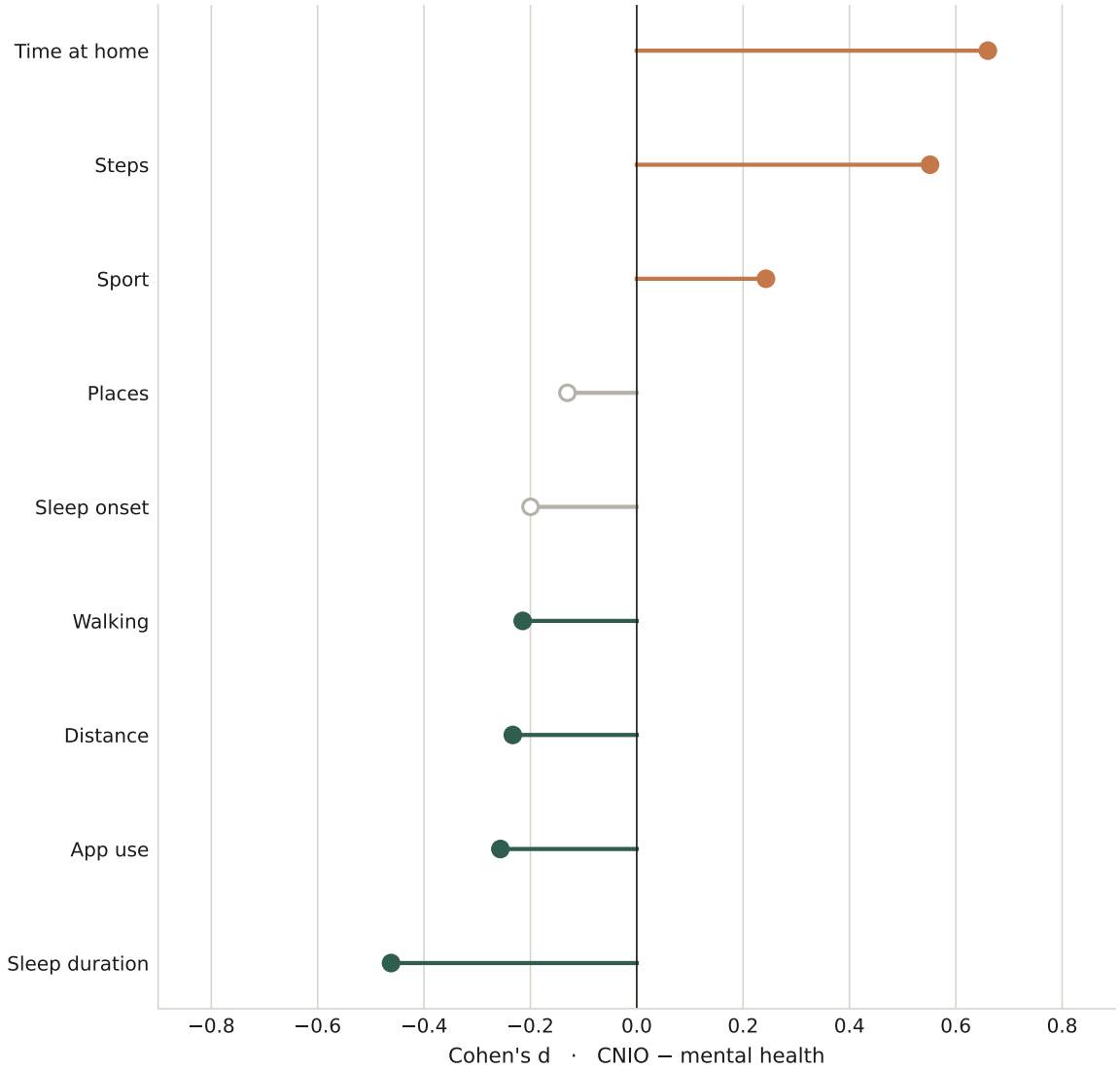
The same topic is the same kind of day

Color is the observed median, centered across the six topics of that cohort. Red is high.



Where the cohorts already differ

Patient-level means. Cohen's d is oncology minus mental health.
Filled markers survive Benjamini–Hochberg FDR at 0.05.



Green: mental health higher. Orange: CNIO higher. Grey: not significant.

Patient means, features that separate

	Sleep	Home	Steps	Sport	App	Walking	Distance
CNIO	8.4 h	15.2 h	7,278	0.47	3.1 h	44 min	23.7 km
Mental	9.3 h	11.0 h	5,432	0.40	3.8 h	57 min	37.9 km

Sleep onset and number of places do not separate (FDR 0.065 and 0.66).

What the figures support

Held out

1,358 mental-health patients, vqvae = 0. The model did not train on them.

Reconstruction

Scaled correlation 0.83 versus 0.77 in CNIO. Sleep onset, about 24 minutes in both.

The code

One of 256 codes, second code farther (margin 0.82 and 0.98). Closer in mental health: distance 4.0 versus 6.2.

No collapse

All 256 codes are used in the held-out cohort. Effective N is 147, against 197 in CNIO.

Same day, other mix

Same topic, similar day in both cohorts. Long-sleep days are 35% in mental health and 9% in CNIO.

A patient is a mixture

64 and 80 codes. The top code covers under 10% of a patient's days.